LaSalle County Station
2601 North 21st Road
Marseilles, IL 61341
815-415-2000 Telephone
www.exeloncorp.com

10 CFR 50.73

RA18-025

April 18, 2018

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating Licenses No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2018-002-00, Damaged Bus Bar Identified Potentially Affecting High Pressure Core Spray System

In accordance with 10 CFR 50.73(a)(2)(v)(D), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2018-002-00 for LaSalle County Station, Unit 1.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

Harold Vinyard

Harold T. Vinyard
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

NRC FORM 366 (02-2018)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
LaSalle County Station, Unit 1	05000373	1 OF 4

4. Title

Damaged Bus Bar Identified Potentially Affecting High Pressure Core Spray System

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name: NA	Docket Number: NA
02	17	18	2018	002	00	04	18	18	Facility Name: NA	Docket Number: NA

9. Operating Mode: 1

10. Power Level: 94

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
John Kowalski, Maintenance Director	(815) 415-2500

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	BG	BU	G080	Yes	NA	NA	NA	NA	NA

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No		NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 17, 2018, during troubleshooting for an issue with the Unit 1 “B” diesel generator (DG) oil circulating pump, station personnel identified damage of a bus bar at the breaker that supplies the DG auxiliaries. One of the loads fed from this breaker is the Division 3 DC Battery Charger. Station personnel determined that the degradation of the bus bar may have prevented the Division 3 DC battery charger from performing its function, which could have prevented the High Pressure Core Spray (HPCS) system, which is a single-train safety system, from performing its design safety function. This condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of the safety function of structures or systems that are required to mitigate the consequences of an accident. Based on a subsequent review of the loads powered by the 2B cubicle, loss of any of those loads would not have prevented the 1B DG from being able to start and perform its design function. The condition of the motor control center (MCC) 143-1 (1AP79E) bus-bar was the result of a degraded connection between the “A” phase bus bar and the “A” phase bus clip (stab) associated with the breaker cubicle 2B. The apparent cause of the degraded bus bar to clip connection is due to relaxation of the bucket clip over time. The condition was exacerbated by long-term continuous ohmic heating caused by the normal currents of the loads supplied by this molded case circuit breaker. Corrective actions included replacement of the degraded clip assembly with one obtained from a spare cubicle, relocating loads to alternative spare cubicles in the affected MCC, visual inspection of all bucket-to-bus connections and repairs as necessary, and performance of a causal investigation including component failure analysis. This model of MCC used at the station is limited to Division 3 on both operating units; therefore, the extent of condition does not apply to other safety-related divisions.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2018	- 002	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The affected component was the Division 3 480-volt motor control center (MCC) 143-1 (1AP79E) that is powered by the Division 3 4160-volt 143 switchgear. This MCC provides power to components that are used to support operation of the High Pressure Core Spray (HPCS) system, which is part of the Emergency Core Cooling System (ECCS). In accident situations, switchgear 143 is powered by the 1B diesel generator (DG). Specifically, the connection of the "A" phase of the 2B cubicle to MCC 143-1 bus bar was discovered degraded. This cubicle feeds Division 3 battery charger and several auxiliary components that support operation of the 1B DG.

CONDITION PRIOR TO EVENT

Unit(s): 1	Date: February 17, 2018	Time:	0200 CST
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level:	94 percent

DESCRIPTION

On February 16, 2018, at approximately 1418 CST the main control room (MCR) received Plant Process Computer (PPC) alarms associated with the Division 3 1B DG, including an alarm for the 1B DG oil pump. Approximately 15 minutes later the 1B DG trouble alarm annunciated, and the local panel alarm for low oil pressure was identified. Field investigations determined the 1B DG oil circulating pump had not been rotating. In accordance with an engineering evaluation, the alternate current (AC) circulating oil pump is not required for 1B DG operability provided oil and water temperatures remain above 85 degrees Fahrenheit (F). A special log was started to verify temperatures remained above 85 degrees F.

On February 16, 2018, at approximately 2129 CST, the 1B DG was placed into maintenance, and the HPCS system was declared inoperable to support troubleshooting associated with the cubicle for 1B DG auxiliaries. During troubleshooting for the failure of the 1B DG Oil Circulating Pump to rotate, the Electrical Maintenance Department (EMD) technicians identified low voltage on one phase of the motor. Specifically, EMD removed the MCC bucket from the bus, and the inspection identified that the "A" phase of the bus had localized damage in the location where the MCC 143-1 cubicle 2B bucket (1AP79E-2B) connects to the bus-bar. The bus bar degradation occurred at the first clip connection point from the top of the bus bar. The damaged area was toward the forward edge (i.e., cubicle side) of the bus bar and extended back approximately 0.557-inches on both sides of the bar. The damaged area was approximately 1-inch long, which is the approximate length of the clip.

On February 17, 2018, at approximately 0200 CST (time of discovery) the station determined that the degradation of the bus bar may have prevented the Division 3 DC battery charger from performing its function, which could have prevented the HPCS system, a single-train safety system, from performing its design safety function. This condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of the safety function of structures or systems that are required to mitigate the consequences of an accident. [Reference ENS 53219.]

<u>Station Response</u>

The "A" phase clip of the sliding disconnect assembly mounted on the bucket was also found to be have degraded clip connection (i.e., relaxation) and determined to have caused the degradation of the "A" phase bus bar. This condition was determined to have resulted in a single-phasing condition which prevented the 1E22-S001-C oil circulating pump from rotating. This MCC cubicle provides power for seven 1B DG components: AC oil circulating pump, DC battery charger, immersion heater, 1B starting air compressor, 1B refrigerated air dryer, generator space heater, and AC soak back oil pump.

A temporary modification was initially implemented to supply temporary power to 1AP79E-2B loads utilizing a spare cubicle in MCC 143-1 (1AP79E). Post-maintenance verifications were performed to ensure proper equipment function was restored. Subsequently, the relocation of this cubicle was made permanent by another modification package.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2018	- 002	- 00

NARRATIVE

Extent of Condition

An extent of condition review was performed on the remaining MCC 143-1 cubicles. Of the 25 cubicles (including spare cubicles) on MCC 143-1, ten discrepancies were noted on six individual cubicles where less than optimal clip to bus bar contact areas were identified. These discrepancies involved either higher than expected resistance readings or gaps in the contact areas of the clip to bus bar connection. All but one of the six cubicles with gap discrepancies involved only the top or bottom of one side of the clip surface contacting the bus bar, resulting in the clip making point contact with the bus bar versus the preferred line contact. Visual inspection of the discrepant connections did not identify any degradation of the bus bar surface area. The discrepant connections were addressed by reforming the clip, followed by visual inspection to ensure the clip to bus bar contact was restored to line contact. Resistance readings of the repaired connections were then obtained.

CAUSE

A causal investigation was performed, after obtaining the results of a failure analysis performed at an off-site testing facility. The investigation determined proximate cause, apparent cause, and contributing causal factor as indicated below.

The cause of the damage to bus 143-1 is the result of a degraded connection between the “A” phase bus bar and the “A” phase bus clip associated with the 2B cubicle. The presence of heating and arcing damage on both the clip and the bus bar indicated that a high resistance connection developed. Based on the lab observations and testing, the cause for the increased resistance was the opening of the clip sides beyond the approximately 3/16-inch width of the bus bar.

The apparent cause of the degraded bus bar to clip connection is due to relaxation of the bucket clip over time. The condition was exacerbated by long term continuous ohmic heating caused by the normal currents of the seven loads supplied by this molded case circuit breaker. This heating caused further degradation leading to the event. This causal factor is supported by failure analysis and evidence obtained during the inspection of the International Switchboard MCC 143-1 which identified less than optimal clip to bus bar interface on six cubicles. Three of the six cubicles with discrepancies were spare cubicles (fifty percent) which have never been subjected to long term heating due to carrying load currents. Additionally, no preventative maintenance (PM) task is performed on the spare cubicles; therefore, plastic deformation from removal and installation is unlikely due to the bucket to bus bar connection not being routinely disturbed.

A contributing causal factor is the absence of a holistic MCC inspection. The previously retired PM activities will be reinstituted on the Unit 1 MCC 143-1 and Unit 2 MCC 243-1 through an action generated from the causal evaluation.

Based on input from craft electricians, the sliding interface on the International Switchboard MCC is not as precise as the interface provided on a Klockner-Moeller MCC that comprises all the 480-Volt AC Division 1 and Division 2 MCCs. Based on this input, it is believed some plastic deformation of the 1AP79E-2B clips could have taken place during previous maintenance. However, no evidence exists to conclusively identify this as a causal factor. As a procedure enhancement, an action will be assigned to add a caution note to the MCC cubicle inspection procedure about the importance of correct bucket to bus bar alignment prior to engaging the primary disconnect clips to the bus bars.

REPORTABILITY AND SAFETY ANALYSIS

This condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. This condition was reported to the NRC via Event Notification System (ENS) report number 53219 on February 17, 2018.

This condition was not a safety system functional failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline." The SSFF evaluation determined that the bus connection clip on the cubicle and location on the bus-bar where the clip was connected were damaged. This was also determined to be the cause of the single-phasing of the HPCS DG oil circulating pump. Based on subsequent review of the loads powered by the 1AP79E-2B cubicle, loss of any of those loads would not have prevented the 1B DG from being able to start and perform its design function.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2018	- 002	- 00

NARRATIVE

CORRECTIVE ACTIONS

Immediate corrective actions taken in response to the condition were:

- Replacement of the degraded clip assembly with one obtained from a spare cubicle
- Relocate all required loads from the degraded section of MCC 143-1 (1AP79E)
- Inspect MCC 1AP79E, including a visual inspection of all bucket-to-bus connections and repair as necessary
- Perform failure analysis for the failed components

A causal investigation was performed, after obtaining the results of a failure analysis performed at an off-site testing facility. The investigation determined additional corrective actions as follows.

- Code Unit 2 MCC 243-1 (2AP79E) inspection work order as corrective action related
- Reinstate previously retired preventative maintenance tasks for MCC cleaning and bus bar inspections for 1(2)AP79E
- Evaluate impact of failure on the Maintenance Rule program
- Develop procedure revision to add caution statement to cubicle inspection procedure LES-GM-108 to identify the potential for plastic deformation of the primary disconnects when reconnecting the bucket to the bus bars
- Additionally, lessons learned from the recent Unit 1 and Unit 2 Division 3 MCC inspections will be incorporated

PREVIOUS OCCURRENCES

There have been no similar events or conditions at the station in the previous three years.

COMPONENT FAILIRE DATA

Manufacturer: General Electric Company (G080)
Device: Motor Control Center (MCC) 143-1 Breaker 1AP79E-2B
Component ID: Model Series 170 International Switchboard (I147)